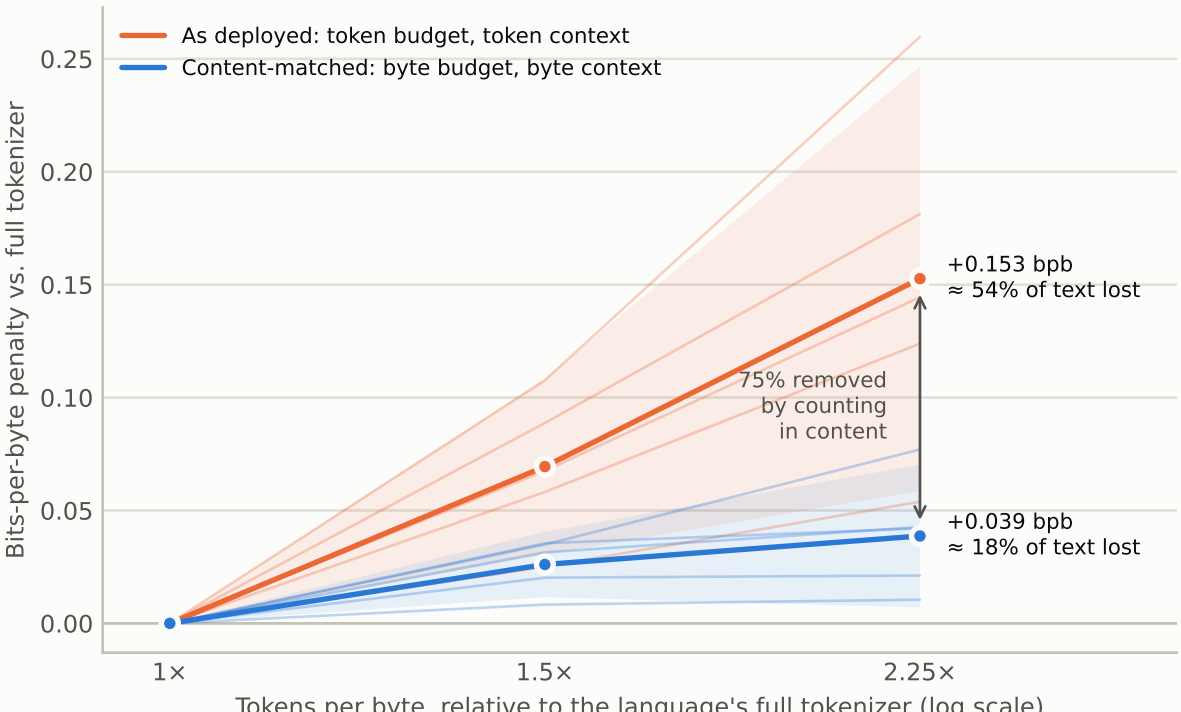
Figure 1. The tokenization tax, pooled over 5 languages (model M)
As deployed: token budget, token context
Content-matched: byte budget, byte context
Bits-per-byte penalty vs. full tokenizer
0.25
0.20
0.15
0.10
0.05
0.00
+0.153 bpb
≈ 54% of text lost
75% removed
by counting
in content
+0.039 bpb
≈ 18% of text lost
1×
1.5×
2.25×
Tokens per byte, relative to the language's full tokenizer (log scale)